

Caputo Dirichlet Heat Flow: Sharp Smoothing, Schatten, and Resolvent Limits

Route-A source-local certificate HCS-C277

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Abstract

We solve the Caputo time-fractional heat equation on $(0, \pi)$ in its exact Dirichlet sine basis and prove inverse-stable subordination, positivity, and contraction for every $0 < \beta \leq 1$. For $0 < \beta < 1$ the solution family is not a semigroup; at every positive time it gains exactly two spatial derivatives and belongs to $\mathcal{S}_p$ exactly when $p > 1/2$. Its scaled long-time limit is the Dirichlet resolvent in operator norm, whereas $\beta = 1$ recovers all-order heat smoothing and exponential decay; 378 executable cells audit every endpoint. No source operator ideal is promoted to target arithmetic data.

1 The frozen memory owner

Let $X = L^2(0, \pi)$ and

$$A = -\frac{d^2}{dx^2}, \quad D(A) = H^2(0, \pi) \cap H_0^1(0, \pi).$$

The normalized eigenvectors $e_n = (2/\pi)^{1/2} \sin(nx)$ satisfy $Ae_n = n^2 e_n$. We freeze

$${}^C D_t^\beta u(t) + Au(t) = 0, \quad u(0) = u_0 \in X, \quad 0 < \beta \leq 1. \quad (1)$$

For $0 < \beta < 1$,

$${}^C D_t^\beta f(t) = \frac{1}{\Gamma(1-\beta)} \int_0^t (t-s)^{-\beta} f'(s) ds,$$

and $\beta = 1$ means the ordinary derivative. Bounded-domain fractional diffusion-wave problems admit eigenfunction constructions of this type [2]; here the interval permits sharp operator thresholds.

Writing

$$E_\beta(z) = \sum_{k=0}^{\infty} \frac{z^k}{\Gamma(\beta k + 1)},$$

the coefficient identity

$${}^C D_t^\beta \frac{t^{\beta k}}{\Gamma(\beta k + 1)} = \frac{t^{\beta(k-1)}}{\Gamma(\beta(k-1) + 1)}, \quad k \geq 1,$$

gives the unique solution family

$$S_\beta(t)u_0 = \sum_{n \geq 1} E_\beta(-n^2 t^\beta) \langle u_0, e_n \rangle e_n. \quad (2)$$

Pollard proved that $E_\beta(-x)$ is completely monotone on $x \geq 0$ for $0 < \beta < 1$ [1]. More precisely, if $\eta_\beta(t, s)$ is the inverse-stable density characterized by

$$\int_0^\infty e^{-zt} \eta_\beta(t, s) dt = z^{\beta-1} e^{-sz^\beta},$$

then scalar Laplace inversion in (2) yields

$$S_\beta(t) = \int_0^\infty e^{-sA} \eta_\beta(t, s) ds. \quad (3)$$

Thus $S_\beta(t)$ is self-adjoint and positive, preserves nonnegative functions, and is contractive. Dominated convergence in (2) gives strong continuity at zero.

2 The sharp memory theorem

We use the spectral Sobolev scale $X^s = D(A^{s/2})$. In particular, $X^2 = D(A) = H^2 \cap H_0^1$; “two derivatives” includes this endpoint.

Theorem 1 (category, smoothing, and Schatten boundaries). *Fix $0 < \beta < 1$ and $t > 0$.*

1. *The strongly continuous family S_β is not a semigroup.*
2. *Within the declared domain $\theta \geq 0$, $A^\theta S_\beta(t)$ is bounded on X if and only if $\theta \leq 1$. Equivalently, within $s \geq 0$, $S_\beta(t) : X \rightarrow X^s$ is bounded if and only if $s \leq 2$.*
3. *For every $p > 0$, $S_\beta(t) \in \mathcal{S}_p$ exactly when $p > 1/2$. The endpoint $p = 1/2$ fails.*

Proof. If S_β were a semigroup, the continuous first-mode multiplier $E_\beta(-t^\beta)$ would be an exponential. But

$$E_\beta(-t^\beta) = 1 - \frac{t^\beta}{\Gamma(1 + \beta)} + O(t^{2\beta}),$$

which is not exponential when $\beta < 1$.

On the positive ray, the standard Mittag-Leffler expansion gives

$$E_\beta(-x) = \frac{1}{x\Gamma(1 - \beta)} + O(x^{-2}) \quad (x \rightarrow \infty). \quad (4)$$

If a later reciprocal-Gamma coefficient vanishes, (4) only improves. Hence

$$n^{2\theta} E_\beta(-n^2 t^\beta) \sim \frac{t^{-\beta}}{\Gamma(1 - \beta)} n^{2\theta - 2}.$$

Continuity on compact x -intervals supplies the upper bound at $0 \leq \theta \leq 1$, and the positive asymptotic proves failure at every $\theta > 1$. If $\theta < 0$, then $A \geq I$ makes A^θ bounded, so $A^\theta S_\beta(t)$ is also bounded; these negative powers lie outside the declared $\theta \geq 0$ smoothing domain. The singular values themselves are asymptotic to a positive constant times n^{-2} , so their p th powers sum exactly for $2p > 1$. $\square$

The endpoint is substantive: memory retains only one bounded power of the second-order generator. This differs from merely asserting some unspecified regularization, and it prevents importing the heat flow’s analytic smoothing into the fractional clock.

3 Long time and the classical face

Theorem 2 (operator-norm resolvent limit). *For $0 < \beta < 1$,*

$$t^\beta S_\beta(t) \longrightarrow \frac{A^{-1}}{\Gamma(1 - \beta)} \quad \text{in } \mathcal{B}(X), \quad (5)$$

and $t^\beta \|S_\beta(t)\| \rightarrow 1/\Gamma(1 - \beta)$. For $\beta = 1$, $S_1(t) = e^{-tA}$, $\|S_1(t)\| = e^{-t}$, $A^\theta S_1(t)$ is bounded for every $\theta \geq 0$, and $S_1(t) \in \mathcal{S}_p$ for every $p > 0$.

Proof. Set $g(x) = xE_\beta(-x) - 1/\Gamma(1 - \beta)$. Equation (4) gives $g(x) \rightarrow 0$. The n th diagonal entry of the difference in (5) is

$$\frac{g(n^2 t^\beta)}{n^2},$$

whose supremum is bounded by $\sup_{x \geq t^\beta} |g(x)| \rightarrow 0$. Complete monotonicity makes the first multiplier the norm, proving the scalar limit. At $\beta = 1$, (2) becomes $e^{-n^2 t}$; exponential mode decay proves all remaining claims. $\square$

The heat face is therefore a singular structural boundary, not a uniform extension of (4): algebraic memory tails disappear, the solution family becomes a semigroup, and finite smoothing saturates to all orders.

4 Executable certificate and Route-A boundary

The receipt contains 24 scalar Mittag–Leffler cells, 192 spectral cells, six composition witnesses, 96 large-time cells using the independent identity $E_{1/2}(-x) = e^{x^2} \operatorname{erfc}(x)$, 35 smoothing cells, and 25 Schatten cells. A producer-independent checker, 90 symbolic identities, fresh byte replay, and repaired-hash mutations audit the package. These finite cells test conventions; (2)–(5) prove the parameter theorem.

Under `NO_BAD_EULER_OR_ROOT_NUMBER`, the tuple is

`(AO_FAIL, A1_FAIL, A2_FAIL, A3_FAIL, A4_FORMAL_HINT),`

overall `ROUTE_A_REJECTED`; Route B is disabled. Trace-ideal membership of this dissipative memory family is not a target determinant, functional equation, divisor, or Hilbert–Pólya operator.

References

- [1] H. Pollard, *The completely monotonic character of the Mittag-Leffler function $E_a(-x)$* , Bull. Amer. Math. Soc. 54 (1948), 1115–1116, [doi:10.1090/S0002-9904-1948-09132-7](https://doi.org/10.1090/S0002-9904-1948-09132-7).
- [2] K. Sakamoto and M. Yamamoto, *Initial value/boundary value problems for fractional diffusion-wave equations and applications to some inverse problems*, J. Math. Anal. Appl. 382 (2011), 426–447, [doi:10.1016/j.jmaa.2011.04.058](https://doi.org/10.1016/j.jmaa.2011.04.058).